

REPOSITORY PLAN

Manuscript: “Institutional Observability Under Scaled AI Governance”

Date: June 3, 2026

Decision: CREATE REPOSITORY

A GitHub repository is **recommended** for this manuscript. The simulation code, figure generation, and supplementary materials are substantive enough to warrant a standalone reproducibility package.

Repository Name

oversight-saturation-sedi

Folder Structure

```
oversight-saturation-sedi/
|----- README.md
|----- LICENSE
|----- CITATION.cff
|----- .gitignore
|----- code/
|   |----- requirements.txt
|   |----- environment.yml
|   |----- monte_carlo_simulation.py      # Main MC simulation (120 runs)
|   |----- kingman_analysis.py           # Kingman heavy-traffic approximation
|   |----- depth_degradation.py          # D(rho) degradation function
|   |----- sedi_computation.py           # SEDI index calculation
|   |----- figure_generation.py          # All figure generation
|   |----- utils.py                      # Shared utilities
|----- figures/
|   |----- fig_saturation.pdf             # Figure 1: Saturation curve
|   |----- fig_divergence.pdf            # Figure 2: O(N)/O(log N) divergence
|   |----- fig_routing.pdf               # Figure 3: Routing architecture
|   |----- fig_depth_kde.pdf             # Figure 4: Review depth KDE
|   |----- fig_sedi.pdf                  # Figure 5: SEDI degradation
|----- data/
|   |----- simulation_outputs/            # CSV outputs from MC runs
|   |   |----- queue_lengths.csv
|   |   |----- review_depths.csv
|   |   |----- sedi_values.csv
|   |----- calibration/                  # Calibration data references
|   |----- README.md                     # Links to Vanderbilt, PHTI sources
|----- supplementary/
```

	-----	appendix_a_derivations.pdf	# Full mathematical derivations
	-----	appendix_b_sensitivity.pdf	# Sensitivity analysis
	-----	appendix_c_scenarios.pdf	# Additional scenarios
	`-----	appendix_d_calibration.pdf	# Calibration methodology
`-----		manuscript/	
	`-----	README.md	# Link to published version

Required Files

1. README.md

- Paper title and authors
- Abstract
- Repository overview
- Dependencies and installation
- How to run simulations
- How to reproduce figures
- Citation information
- License information

2. LICENSE

- MIT License (permissive, standard for academic code)
- Enables maximum reuse while requiring attribution

3. CITATION.cff

```
cff-version: 1.2.0
message: "If you use this software, please cite it as below."
authors:
  - family-names: "[AUTHOR]"
    given-names: "[AUTHOR]"
title: "Oversight Saturation: Simulation Code for SEDI Index"
version: 1.0.0
date-released: 2026-06-03
url: "https://github.com/[username]/oversight-saturation-sedi"
```

4. requirements.txt

```
numpy>=1.24.0
scipy>=1.10.0
matplotlib>=3.7.0
pandas>=2.0.0
seaborn>=0.12.0
```

5. .gitignore

```
__pycache__/*
*.pyc
```

```
.ipynb_checkpoints/  
.DS_Store  
*.aux  
*.log  
*.out
```

Reproducibility Package

The repository serves as a complete reproducibility package: - All code to regenerate every figure and table - All simulation parameters documented - All outputs archived - Clear instructions for independent verification

Hosting

- **Primary:** GitHub (public repository)
 - **Archive:** Zenodo (DOI-minted snapshot for permanence)
 - The Zenodo DOI should be cited in the manuscript's data/code availability statement
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Post-Submission Workflow

1. Create GitHub repository (private during review)
2. Push all code and supplementary materials
3. Mint Zenodo DOI after acceptance
4. Update manuscript with DOI before publication
5. Make repository public upon publication